

Scalability & Future Plan

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical weather data only	Prediction accuracy may decrease during unexpected events	HIGH
2	No real-time weather API integration	Users must manually enter weather parameters	HIGH
3	Limited deployment and scalability	The current Flask application is suitable for small-scale	MEDIUM

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud platforms such as IBM Cloud
2	Data Storage	Uses a local dataset and stores	Integrate cloud databases and real-time weather data
3	Performance	Predictions are generated locally	Optimize model performance, enable caching
4	Security	Basic input validation	Implement user authentication, HTTPS, API security

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2			
Phase 3			
Phase 4			